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II/IV-B.Tech(EVEN Sem), Academic Year: 2025-2026
B.Tech(CSE, CSIT &ME), 2024 Batch II/IV, EVEN Semester
Subject Code: 24MT2019 / TITLE: Probability and Statistics
Home Assignment-1 (CO-1)

1. In a two-week study of the productivity of workers, the following data were obtained on the total number of acceptable pieces which 100 workers produced?

65	36	49	84	79	56	28	43	67	36
43	78	37	40	68	72	55	62	22	82
88	50	60	56	57	46	39	57	73	65
59	48	76	74	70	51	40	75	56	45
35	62	52	63	32	80	64	53	74	34
76	60	48	55	51	54	45	44	35	51
21	35	61	45	33	61	77	60	85	68
45	53	34	67	42	69	52	68	52	47
63	65	55	61	73	50	53	59	41	54
41	74	82	58	26	35	47	50	38	70

Group these figures into a distribution having the classes 20-29, 30-39, 40-49,…… and 80-89, and plot a histogram and frequency polygon using *20, 30),…..*80, 90); where the left hand end point is included but the right hand end point is not.

2. Material manufactured continuously before being cut and wound into large rolls must be monitored for thickness (caliper). A sample of 10 measurements on paper, in millimeters, yielded:

   32.2, 32.0, 30.4, 31.0, 31.2, 31.2, 30.3, 29.6, 30.5, 30.7

   Find (a) Mean (b) Median (c) Mode

3. During the manufacture of hard disks, the height between the disk and the head, the head lift, must be controlled. One manufacturer recorded:

   0.239, 0.246, 0.245, 0.243, 0.239, 0.241, 0.248, 0.246, 0.243, 0.242, 0.251, 0.246

   Find the range; mean deviation; variance; standard deviation and coefficient of variation. Measurements from a large hard disk have sample mean 0.280 and standard deviation 0.05. Which is relatively more variable?

4. A.A. Michelson (1852-1931) made many series of measurements of the speed of light. Using a revolving mirror technique, he obtained:
12, 30, 30, 27, 30, 39, 18, 27, 48, 24, 18, for the differences (velocity of light in air-(229,700) km/s). Find the Range; Mean Deviation; Quartile Deviation, Variance and Standard Deviation. Also obtain Coefficient of Variation.

5. An analysis of monthly wages paid to the workers of two firms A and B belonging to the same industry gave the following results :

	Firm-A	**Firm-B**
Number of Wage-earners	986	548
Average hourly wages	Rs.52.5	Rs.47.5
Variance of distribution of wages (Rs.^{2})	100	121

i) Which firm, A or B, pays out larger amount as hourly wages?

ii) In which firm, A or B is there greater variability in individual wages?

6. a) The following table shows the distribution of the number of students per teacher in 750 colleges :

Students	1	4	7	10	13	16	19	22	25	28
Frequency	7	46	165	195	189	89	28	19	9	3

Draw the histogram for the data and superimpose on it the frequency polygon.

b) Draw a histogram for the following data:

Age (in years)	2-5	5-11	11-12	12-14	14-15	15-16
No. of Boys	6	6	2	5	1	3

c) The adjoining data relate to the ages of a group of workers in a factory and number of workers. Draw less than and more than ogives for the data given below:

Ages	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
No. of Workers	35	45	70	105	90	74	51	30

7. Obtain Mean, Median and Mode from the data given below :

Weight(Kg.)	93-97	98-102	103-107	108-112	113-117	118-122	123-127	128-132
No. of students	2	5	12	17	14	6	3	1

8. Calculate Mean and Standard deviation for the following table which gives the age distribution of 542 members by applying
i) Assumed Mean Method and ii) Step-deviation Method.

Age(in years)	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of members	3	61	132	153	140	51	2

9. a) Obtain the Five-point summary of the following data (i.e., to calculate, Range, Median, Mean Deviation, Quartile Deviation and Inter-Quartile.

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	5	8	12	15	20	14	12	6

b) Obtain the Five-point summary of the following data, which reflects the waiting time of an individual to wait for a local train in a Metropolitan city (In Seconds).

82	45	64	80	82	74	79	80	80	78	80	80	48	73	80	79	81	70	78	73

10. Lives of two models of refrigerators turned in for new models in a recent Survey are given in the adjoining table. What is the average life of each model of these refrigerators? Which model shows more uniformity?

Life (No. of years)	**Model-A**	**Model-B**
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	9
10-12	4	1